

ANGRA DO HEROISMO, Portugal

WMO: 085110

Lat: 38.6600N

Lon: 27.2242W

Elev: 91

StdP: 100.24

Time Zone: -1.00 (AZR)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	9.1	10.4	3.6	5.0	10.6	4.6	5.3	11.4	13.1	14.6	11.7	14.2	3.7	310	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.8	25.7	21.9	25.1	21.5	24.4	21.1	23.3	24.5	22.7	24.0	22.1	23.6	3.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.9	17.9	23.9	22.2	17.0	23.6	21.5	16.4	23.2	69.5	24.5	67.1	24.2	65.0	23.7	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
10.6	9.2	8.0	DB	6.8	27.4	5.4	0.8	2.9	27.9	-0.3	28.4	-3.3	28.8	-7.2	29.4
			WB	5.9	23.7	1.0	0.5	5.2	24.0	4.6	24.3	4.1	24.6	3.3	24.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	14.9	14.5	14.5	14.7	16.7	18.9	21.3	22.4	21.4	19.2	16.9	15.4
	DBStd	3.26	1.56	1.61	1.55	1.65	1.43	1.56	1.42	1.20	1.45	1.79	1.84	1.80
	HDD10.0	1	0	0	0	1	0	0	0	0	0	0	0	0
	HDD18.3	662	107	107	119	110	54	11	0	1	11	50	92	
	CDD10.0	2768	151	127	139	141	206	267	351	386	342	286	206	167
	CDD18.3	388	0	0	0	0	3	28	93	127	92	38	6	1
	CDH23.3	462	0	0	0	0	0	7	100	252	97	6	0	0
	CDH26.7	3	0	0	0	0	0	0	1	2	1	0	0	0
Wind	WSAvg	3.6	4.1	4.1	4.3	4.0	3.4	3.1	2.7	2.8	3.1	3.7	3.7	4.3
Precipitation	PrecAvg	1047	108	97	119	75	72	46	31	42	78	119	99	152
	PrecMax	1510	199	226	285	211	251	108	77	163	243	333	218	331
	PrecMin	725	24	15	26	20	5	4	8	2	7	29	16	44
	PrecStd	178	43	57	68	57	56	26	20	33	51	71	49	80
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.7	19.1	18.8	19.7	21.5	24.2	26.4	26.9	26.1	24.2	21.6	19.7
		MCWB	17.0	16.2	16.4	16.0	18.3	20.0	22.0	22.2	22.3	20.9	19.3	18.1
	2%	DB	18.1	18.0	17.9	18.5	20.4	23.0	25.3	26.0	25.2	23.1	20.7	19.0
		MCWB	16.6	15.9	16.1	15.6	17.6	20.0	21.6	22.2	21.5	20.3	18.7	17.5
	5%	DB	17.6	17.3	17.2	17.7	19.6	22.2	24.4	25.3	24.5	22.5	20.0	18.5
		MCWB	16.3	15.5	15.3	15.2	17.0	19.5	21.1	21.7	21.1	20.0	18.1	17.1
	10%	DB	17.2	16.7	16.7	17.0	18.9	21.4	23.7	24.7	23.8	21.7	19.4	18.0
		MCWB	15.9	15.1	14.9	14.9	16.6	18.8	20.7	21.3	20.7	19.6	17.7	16.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.2	17.6	17.6	17.2	19.5	21.8	23.9	24.2	23.7	22.1	20.4	19.0
		MCDB	18.3	18.0	18.0	18.1	20.4	22.6	24.7	25.3	24.7	23.0	21.1	19.2
	2%	WB	17.2	16.7	16.7	16.5	18.6	20.9	23.1	23.4	22.8	21.2	19.6	18.1
		MCDB	17.6	17.3	17.3	17.5	19.6	22.2	24.2	24.7	24.0	22.2	20.1	18.7
	5%	WB	16.7	16.1	16.1	16.0	17.9	20.1	22.3	22.8	22.2	20.7	18.9	17.6
		MCDB	17.3	16.8	16.7	16.9	19.0	21.5	23.7	24.3	23.5	21.8	19.5	18.2
	10%	WB	16.2	15.6	15.5	15.5	17.2	19.4	21.5	22.2	21.6	20.2	18.2	17.0
		MCDB	16.9	16.4	16.4	16.6	18.5	20.9	23.1	24.0	23.1	21.4	19.2	17.7
Mean Daily Temperature Range	5% DB	MDBR	3.3	3.9	3.7	4.1	4.1	4.3	4.7	4.8	4.5	4.0	3.5	3.5
		MCDBR	3.3	4.2	3.8	4.4	4.5	4.8	5.1	5.1	4.9	4.3	3.5	3.4
		MCWBR	3.3	3.7	3.3	3.5	3.2	3.3	3.3	3.3	3.4	3.2	3.2	3.4
	5% WB	MCDBR	3.1	3.9	3.4	3.8	4.0	4.3	4.4	4.5	4.1	3.8	3.5	3.0
		MCWBR	3.5	3.9	3.5	3.5	3.2	3.2	3.3	3.4	3.5	3.4	3.7	3.5
Clear-Sky Solar Irradiance	taub	0.361	0.370	0.400	0.398	0.389	0.389	0.381	0.374	0.373	0.379	0.362	0.363	
	taud	2.454	2.438	2.343	2.378	2.439	2.472	2.509	2.550	2.552	2.505	2.483	2.427	
	Ebn at Noon	806	851	857	881	892	889	893	894	876	832	799	771	
	Edh at Noon	83	96	116	118	113	110	105	98	93	89	80	80	
All-Sky Solar Radiation	RadAvg	2.02	2.87	4.01	5.11	5.97	6.32	6.42	6.10	4.83	3.32	2.27	1.71	
	RadStd	0.16	0.16	0.28	0.43	0.41	0.44	0.36	0.24	0.25	0.17	0.12	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	+0.57	N/A	+0.28	+0.43	N/A	N/A	N/A	N/A

Nomenclature: See separate page